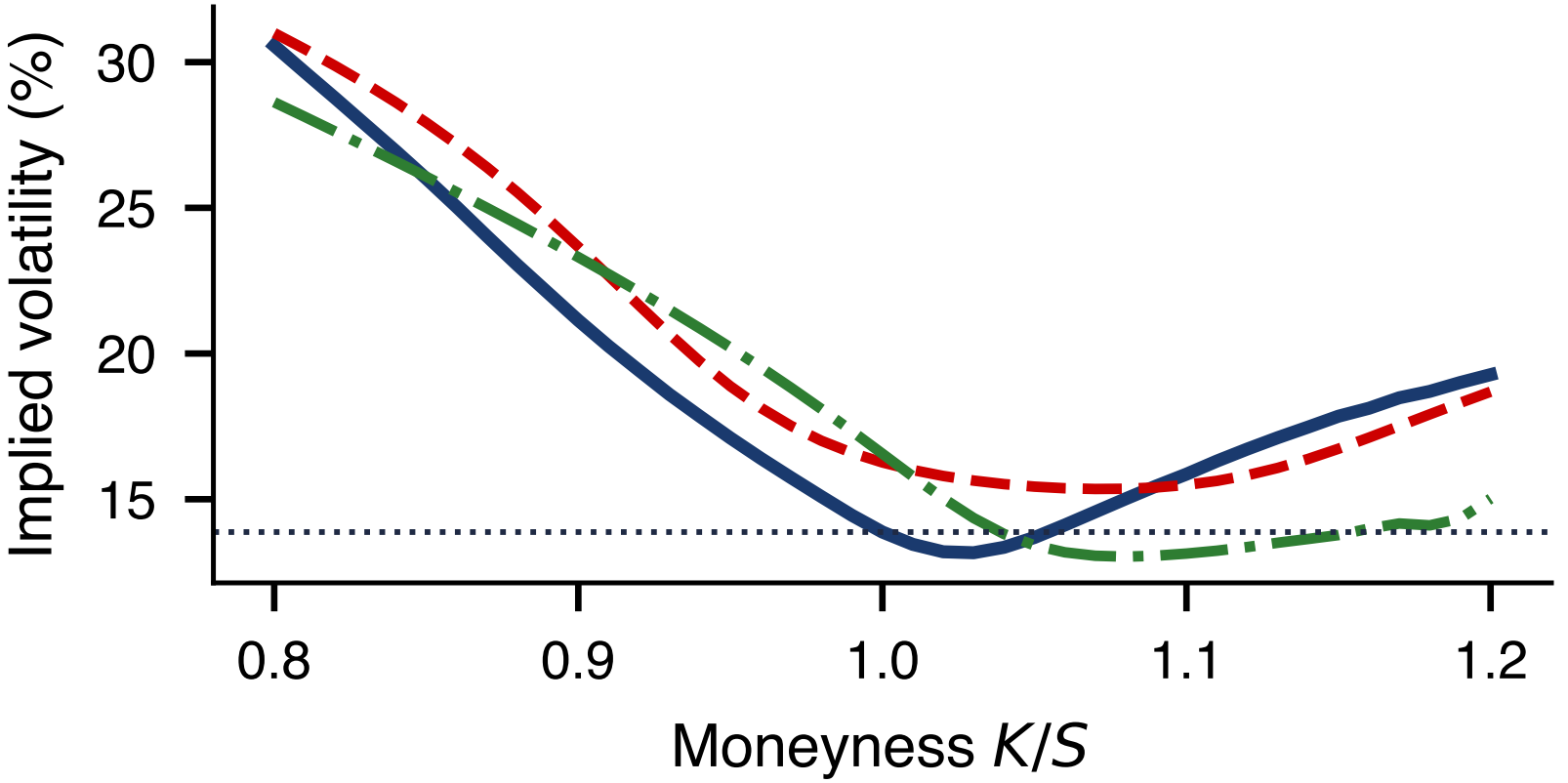

Historical S&P 500 returns (1990-2026)

Merton: one -8% mean crash per year

Heston stochastic volatility, $\rho = -0.7$

Black-Scholes: constant volatility